DA User Guide

DA (short for *Digital Archive*) puts your photos and scanned documents in order. You copy them into one folder, and DA:

- sets exact duplicates aside, so you keep one copy of each photo,
- works out when each photo was taken, and renames it by that date (`photo_19980704_101500_ab12.jpg` was taken on 4 July 1998 at 10:15:00),
- files it in a folder for its year and month (`photos\organized\1998\1998-07\`).

DA never deletes anything. It asks before it moves anything, and it keeps a record of every change, so every step can be checked and most can be undone.

DA runs on Windows 10 and 11.

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Words you'll see

Word	What it means
Archive	A folder DA looks after, for example `D:\jane-archive-199x`. You can have as many as you like, for example one for each person and each decade.
Intake	The folder inside an archive where you put new photos and documents: `intake\`. DA takes them from here.
Organized	Where DA files them: `photos\organized\` and `documents\organized\`, one folder per year and month.
Duplicate	An exact copy of a photo, byte for byte. DA keeps one and moves the others to `_duplicates-holding\`.
Sidecar	A small note DA keeps next to each file, named like the file plus `.meta.json`. It holds the date DA found, a fingerprint of the file, and any tags. Leave sidecars where they are.
Terminal	A window where you type commands. Windows calls it *PowerShell* or *Terminal*. DA's menu runs in one.
Command	Something you type in the terminal, like `da status`, then press **Enter**.

Getting started

1. Get DA

- **Download** DA.exe (the latest version), or get the file from whoever set DA up for you (for example on a USB stick).
- Your browser may ask whether to keep the file. Choose **Keep**.

2. Install it (once per computer)

1. Open **File Explorer** and go to the folder with `DA.exe` (usually **Downloads**).
2. Click in the **address bar** at the top of the window, type `powershell`, and press **Enter**. A terminal window opens in that folder.
3. Type this and press **Enter**: `powershell .\DA.exe install`
4. If Windows shows "*Windows protected your PC*", click **More info**, then **Run anyway**.

5. **Close the terminal window.** From now on, `da` works in any new terminal window.

DA is now at `C:\Scripts\DA\DA.exe`. To update DA later, download the new `DA.exe` and do these steps again; your archives aren't touched.

Windows won't run DA.exe? If you see "*an application control policy has blocked this file*", or Windows Security keeps warning about `DA.exe`, your computer has **Smart App Control** turned on. It only runs programs that are digitally signed, and DA.exe isn't. Don't turn your protection off: install DA with Python instead. It takes about ten minutes, once.

Install DA with Python (if Windows blocks DA.exe)

What's Python? A free, widely used program that runs programs written in the Python language, which DA is. Python itself is signed by the people who make it, so Smart App Control lets it run, and Python runs DA. You don't need to learn any Python: you install it once, type three lines, and then use DA exactly as described in this guide.

A. Install Python (once per computer)

1. Go to python.org, point at **Downloads**, and click the yellow **Download Python** button. Your browser saves the installer (usually in **Downloads**).
2. Open it (double-click it, or click it in the browser's downloads list).
3. Follow what it shows:
   - If you see a window with an **Add python.exe to PATH** box at the bottom, tick it, then click **Install Now**.
   - If a black window asks questions instead (the newer **Python install manager**), type **y** and press **Enter** each time, until it says it's done.
4. When it's finished, close the installer.
5. **Check it:** open a new terminal (press the **Windows** key, type `powershell`, press **Enter**), type this and press **Enter**: `powershell py --version` It answers with something like `Python 3.14.7`. If it says `py` isn't recognized, close the terminal, open a new one, and try again; if it still isn't, run the installer again.

B. Install DA (once per computer)

In a terminal, type these three lines, pressing **Enter** after each, and wait for each to finish (the second one downloads DA and the pieces it needs; it takes a minute or two and prints a lot):

```
py -m venv C:\Scripts\DA\python
C:\Scripts\DA\python\Scripts\python.exe -m pip install -r https://da.gamenightbymike.com/pythor
C:\Scripts\DA\python\Scripts\python.exe -m mindit install
```

What they do: 1. Make a private copy of Python just for DA, in `C:\Scripts\DA\python`. Nothing else on your computer is changed. 2. Download DA (from this website) and what it needs into that copy. 3. Set up the `da`

command and update every archive's `Open DA.cmd` to use it. (If an old `DA.exe` is in `C:\Scripts\DA`, it's renamed `DA.exe.off`, so it doesn't get in the way.)

Close the terminal and open a new one. Type `da version`: it shows DA's version. From here on, everything in this guide works the same: `da`, `Open DA.cmd`, the menu.

C. Update DA later: type `da update`. It downloads the newest version and says which one you have now. If it says it couldn't finish, close every DA window and run it again.

3. Create your first archive (once per archive)

1. Open a new terminal: press the **Windows key**, type `terminal`, press **Enter**.
2. Type this (with your own drive, name and years) and press **Enter**: `powershell da setup D:\jane-archive-199x --years 1990-1999`

What happens depends on the folder:

- **A new folder:** DA creates it, with everything inside it that an archive needs.
- **A folder you've already been copying photos into:** DA lists what's there and asks "*Make it an archive and move these N items into intake\?*". Type **y**. Everything moves into `intake\` with its names unchanged, and the move is written down so it can be undone.
- **A folder that's already an archive:** DA asks "*Use the existing archive?*", keeps its settings, and adds anything missing.

`--years 1990-1999` is optional. With it, a photo taken in another decade (say 2005) is left in `intake\` and listed for you, so you can move it to the right archive (see da split).

Your archive now looks like this:

```
D:\jane-archive-199x\
  Open DA.cmd           <- double-click this to open DA in this archive
  da-archive.yaml       <- this archive's settings
  intake\
    photos\             <- put photos here (any subfolders, any names)
    docs\               <- documents: PDF, Word, Excel
    scans\              <- what your scanner saves: PDFs and scanned pictures
  photos\organized\     <- DA files photos here, by year and month
  documents\organized\  <- and documents here
  output\               <- DA's records and reports
```

4. Open it

Double-click `Open DA.cmd` in the archive folder. DA's menu opens. That's all you need from here on.

The menu

```
DA - jane-archive-199x   D:\jane-archive-199x   years: 1990-1999
intake: 1,204 files waiting    organized: 5,873 photos, 12 documents    last run: 2026-10-13 2

 intake: 1,204 files (from the index)
      1,180  phone-2024
         24  scans
 last run: 2026-10-13T21:31:05  filed 5,873

 1 Process intake    2 Status    3 What needs review    4 Undo last set-aside
 5 Refresh index     6 Switch archive    7 New archive
 L Location folders  S Split by years    M Merge from another copy    R Report (PDF)
 D Fix dates    U User guide    H Help    Q Quit
```

The menu opens with the archive's **Status**: what's waiting in intake, folder by folder, and the last run. It shows it again at the end of **1 Process intake**, and when you switch to another archive. Type the number or letter, then press **Enter**.

Key	What it does
1 Process intake	Files what's in `intake\`. See da run.
2 Status	This archive at a glance: what's waiting, what's filed, the last run.
3 What needs review	What the last run left for you to look at, with the reasons.
4 Undo last set-aside	Puts the duplicates from the last run back where they were.
5 Refresh index	Updates DA's list of files, for example after you moved files yourself.
6 Switch archive	Open another archive on this computer.
7 New archive	Lists the archives this computer knows, then creates a new one (or turns a folder into one).
L Location folders	Groups a month's GPS photos into folders by place. See Group a month into location folders.
S Split by years	Shows which files in intake belong to another decade's archive, and moves them there. See da split.
M Merge from another copy	Copies in what another copy of this archive has and this one doesn't. See Merge two copies of an archive.
R Report (PDF)	Makes the one-page report and opens it.
D Fix dates	Give a date to photos that have none (a folder at a time), then file them. See Fix dates.
U User guide	Opens this guide in your web browser.
H Help	A short reminder of how DA works.
Q Quit	Leaves the menu. The window stays open for typing commands.

Typing `da` on its own in any terminal opens the menu too. Outside an archive, it lists the archives on this computer for you to pick from.

Common workflows

Workflow	What it's for	Menu
Add photos from a phone or camera	Get photos into intake	(File Explorer)
Add scanned documents	Get paper into the archive	(your scanner)
da run	File what's in intake	**1**
da report	See what's there and what needs you	**2**, **3**
Fix what needs review	Help DA with photos it couldn't date	**3**
Duplicates: put back or delete	Undo a set-aside, or free the space	**4**
da split	Move photos from another decade to the right archive	`da split`
Contact sheets	See a whole folder of photos on one page	(made for you)
da find	Search by month, name or tag	steps below
da tag	Label files (`beach` , `grandma` , `taxes`)	steps below
Switch between archives	Work in another archive	**6**
Update DA	Get the newest version	steps below

`da run` is a command you can type. The other "da ..." names stand for a few commands or menu items used together.

Add photos from a phone, camera or memory card

1. **Open the archive's intake folder** in File Explorer: `D:\jane-archive-199x\intake\photos\` .
2. **Make a folder for this batch,** named so you'll recognise it, e.g. `phone-2024-summer` .
3. **Copy the photos into it:**
    - **iPhone:** connect it with its cable and unlock it. Tap **Trust** if it asks. In File Explorer, open **This PC > Apple iPhone > Internal Storage > DCIM** and copy the folders you want. (See iPhone HEIC photos about the photo format.)

    - **Android phone:** connect it, choose **File transfer** on the phone, then open **This PC > (your phone) > Internal storage > DCIM > Camera**.
    - **Camera or memory card:** put the card in the computer (or connect the camera), then open its drive and the **DCIM** folder.
4. **Copy, don't move,** until DA has filed them and you've checked the result.
5. Then process them.

Any folder names and file names are fine. DA works out the dates itself.

Add scanned documents

1. **Set your scanner to save into** `D:\jane-archive-199x\intake\scans\` (or copy the files there afterwards).
2. **PDF is best** for documents. Pictures (`.jpg` , `.png` , `.tif`) saved in `scans\` are treated as documents too, not as photos.
3. Word and Excel files and other PDFs go in `intake\docs\` .
4. Then process them. Documents are filed in `documents\organized\` by the date inside them (when they were scanned or created).

da run: file new photos and documents

Menu: 1 Process intake. DA lists the folders in `intake\` and how many files each holds:

```
intake\photos\
 #  folder                          files
 0  process all of intake\photos\  5,081
 1  phone-2024-summer\             1,204
 2  scans-from-grandma\              877
  number: open   0: process   text: filter   .. up   Enter: back
```

- Type a **number** to open that folder. A folder with no folders inside it is processed straight away.
- Type **0** to process the folder you're in, everything in it.
- Type some **text** (e.g. `grandma`) to show only the folders whose names contain it.
- Type **..** to go up a level, or just press **Enter** to go back.

DA then works through seven steps. Before each step that changes anything, it asks; type **y** to go ahead, or **n** to stop there (nothing after that point changes).

Step	What happens	Changes files?
1 Index	Lists everything in the archive	No
2 Hash	Fingerprints the photos that could be copies	No
3 Duplicates	Moves exact copies to `_duplicates-holding\`	**Yes, if you say y**
4 Preview	Works out each photo's date, new name and folder	Only writes the sidecar notes
5 Check	Shows how many will be filed, by year, and warns about anything odd	No
6 File	Renames and moves the photos into `organized`, then checks each one arrived intact, and makes each folder's contact sheet	**Yes, if you say y**
7 Refresh	Updates the index	No

How long it takes: a few minutes for a few hundred photos; roughly an hour for every 50,000. It's fine to leave it running.

What stays in intake, on purpose: - **Photos with no date anywhere** (none inside the file, none in its name). See Fix what needs review. - **Photos from another decade** (with `years` set). See da split. - **Videos** (`.mov` , `.mp4`): DA doesn't file videos yet. The video half of an iPhone Live Photo is the exception: it's filed with its photo. - **Camera raw files** (`.CR2` , `.NEF` , ...) without a matching JPG. With a matching JPG, the raw file is filed right next to it.

Running it again is always safe. Files already filed aren't touched, and new ones are processed the same way.

From a terminal (handy for leaving a big run going overnight):

```
da run                                  # all of intake
da run intake\photos\phone-2024-summer  # one folder
da run --yes                            # don't ask; go straight through
da run E:\old-backup                    # a folder outside the archive: DA offers to copy it in
```

da report: what's in the archive and what needs you

Menu: 2 Status and **3 What needs review**, or `da report` for a page to print or keep.

A one-page report: type `da report` (add `--open` to open it straight away). DA makes a dated PDF, `output\jane-archive-199x-report-2026-10-14.pdf`, with: - what's filed (photos, documents, space used, the years they span), and how many by year - what's waiting for you (intake, duplicates set aside, notes to review) and the last run - the cameras used, with photo counts, and who took them (from da tag) - the places: how many photos have GPS, a sentence like "*This archive includes photos taken in 2 countries (United States, France) and 9 US states (Illinois, Florida, ...)*", and the most photographed towns (from da location lookup)

The numbers come from DA's index. If you moved files yourself, use `da report --refresh`. A second report on the same day replaces the first.

1. **2 Status:** the archive's name and years, what's waiting in intake, what's filed, the last run.
2. **3 What needs review:** what the last run left behind, grouped by reason, with examples:
    - *No date (in the file or its name)*
    - *Taken outside this archive's years*
    - *File name and camera date disagree* (DA used the date in the name)
    - *Duplicate inside the folder* (left for you to decide)

    The full list is saved as `review.csv` in that run's folder (DA shows where). It opens in Excel. 3. **More detail, from a terminal:** `powershell da db stats # how many photos and documents, by year da intake status # how far each intake folder has got da validate photos\organized # check what's already filed: names, folders, notes (nothing moves)` 4. **Every run's record** is kept in `output\runs\<date-time>-run\`: a summary, the duplicate list, and the preview and filing reports, file by file.

If you added or moved files yourself, choose **5 Refresh index** first so the numbers are current.

Fix what needs review

Choose **3 What needs review**, then for each reason:

- **No date:** DA couldn't find when the photo was taken. If you know, even roughly, give it one: see Fix dates (menu **D**).
- **Outside this archive's years:** it belongs in another archive. See da split.
- **Name and camera date disagree:** usually the camera's clock was wrong. DA used the date in the name. If that's the wrong one, set the right date (or shift the camera's clock): see Fix dates.
- **Duplicate inside the folder:** two identical files. DA leaves both; delete the one you don't want.

Fix dates: photos with no date, or the wrong one

Scans of old prints, photos from a camera whose clock was never set, files that lost their date when they were copied: DA can't tell when they were taken, so they wait in intake. You can tell it. A date you give is kept with the

photo (in its sidecar) and always wins over the camera and the file name.

The easy way (menu D Fix dates): 1. Choose **D**. DA lists each intake folder that has photos with no date, one at a time: `For each folder, type the date its photos were taken: 1998, 1998-07 or 1998-07-04. scans\album-1 (48, e.g. scan0001.jpg): 1998-07 scans\album-2 (31, e.g. scan0049.jpg): 1999 phone-old (12, e.g. IMG_0003.JPG):` 2. Type a date for a folder, press **Enter** to skip one, or **q** to stop. 3. DA asks whether to process intake now. Say yes, and they're filed by the dates you gave.

Dates you can type:

You know	Type	Filed under
The day	`1998-07-04`	`1998\1998-07\photo_19980704_000000_xxxx.jpg`
The day and time	`1998-07-04 14:30`	`1998\1998-07\photo_19980704_143000_xxxx.jpg`
The month	`1998-07`	`1998\1998-07\` (as the 1st)
Only the year	`1998`	`1998\1998-01\` (January); a month is better if you know it

From a terminal (also for photos already filed): - See which photos have no date: `da date list` (or `da date list intake\scans`). - Give a folder (or photos) a date: `da date set intake\scans\album-1 --date 1998-07`. - **A camera's clock was off** (an hour after a time change, or a year that was never set): shift all its photos by the same amount: `da date shift intake\phone-old --by +1h` (or `-1y`, `+2d`, `-30m`, `+3mo`). - Changed your mind? `da date clear <photos or folder>`: the photo's own date counts again.

For photos already in `organized`, DA then says how to move them to match: `da validate "<their folder>" --mode repair` renames and moves them by the new date.

Duplicates: put back or delete

Exact copies are moved to `_duplicates-holding\<date>\` in the archive, in the same folders they came from.

- **Put them back: 4 Undo last set-aside**. Or, for an earlier run, from a terminal: `da intake undo "<run folder>\set-aside-moved.csv"`.
- **Delete them** to free the space, once you're happy with the result: delete the `_duplicates-holding` folder. Every file in it is an exact copy of one that was kept.

da split: move files to the right decade's archive

An archive with `years` set (e.g. `1990-1999`) only files photos taken in those years. One taken in another decade stays in intake, listed under **3 What needs review** as "*Taken outside this archive's years*". `da split` moves those to the archive for their decade. You don't type the years: each archive's own `years` setting decides.

1. **See where they belong.** In the archive whose intake holds them (e.g. `D:\jane-archive-199x`), type `da split`. DA works out each file's date the same way `da run` does and says: `14 files in intake were taken outside 1990-1999. 11 belong in jane-archive-200x (2000-2009): da split --to "D:\jane-archive-200x" 3 from the 1980s have no archive on this PC yet. Make one: da setup <folder> --years 1980-1989`
2. **No archive for a decade yet?** Make one: `da setup D:\jane-archive-198x --years 1980-1989`.
3. **Move them:** `da split --to D:\jane-archive-200x`. DA lists what will move and asks first. Each photo goes with its sidecar, raw file and Live Photo video into the other archive's `intake\photos\from-jane-archive-199x\`, keeping the folders it was in. Documents go to its `intake\docs\`. A file whose name is already there stays put.
4. **File them there:** open the other archive (its `Open DA.cmd`, or **6 Switch archive**) and choose **1 Process intake**.

Good to know: - **Changed your mind?** `da split` prints an **Undo** line: `da intake undo "<the file it names>"` brings everything back. It works between drives too. - **Only some years:** `da split --to D:\jane-archive-200x --years 2000-2004`. - **An archive that takes any year** (no `years:`) has nothing "outside" it. To move some photos anyway, give `--years`. - Files with no sure date stay in intake. Fix their dates first (see Fix what needs review), then run `da split` again.

da find: search the archive

DA keeps an index of every file: its name, folder, date and tags. Choose **5 Refresh index** first if anything changed, then in a terminal in the archive:

- **Everything from one month** (by the folder it's filed in): `powershell da db query "SELECT file_name, folder FROM files WHERE area='photos' AND folder LIKE '2010\2010-07%'"`
- **By name:** `... WHERE file_name LIKE '%0704%'`. For the name a file had before DA renamed it: `... WHERE source_path LIKE '%Beach%'`.
- **Documents:** use `area='documents'` instead of `area='photos'`.
- **By tag** (see da tag): `da findit tags photos\organized -r beach -v`. Add `-r grandma` for both tags, `-x beach` to leave a tag out, or `--date-prefix 2010-07` for one month. Automatic tags work the same way: `-r owner:jane -r year:2021` finds Jane's photos from 2021, `-r gps` the ones with a position.

Or just browse: every photo is in `photos\organized\<year>\<year>-<month>\`, in date order, and each folder's `_contact-sheet.pdf` shows them all on a page or two.

Contact sheets: a folder at a glance

Every folder DA files photos into gets a **contact sheet**: `_contact-sheet.pdf`, small pictures of all its photos, 20 to a page, each with its name and the date and time it was taken. Double-click it to see a whole month at once, or print it.

- **It keeps itself up to date.** When da run files more photos into a folder, its sheet is made again. If you add, remove or replace photos yourself, Check what's already filed (`da validate`) makes it again.
- **For folders filed before DA made sheets,** or any other folder of photos (even outside an archive), make them yourself; this does every folder inside the one you name: `powershell da contact-sheets photos\organized\1998` Add `--force` to make every sheet again.
- DA never files, indexes or checks the sheet itself, and deleting one does no harm.

da tag: label photos and documents

A tag is a word attached to a photo (`beach` , `grandma` , `trip:big-bear`), so you can find it later with da find. DA makes some tags for you, and you add your own.

Automatic tags. When DA first files a photo, it notes what the photo says about itself: the camera's maker and model, the lens, the camera's serial number, any owner name the camera recorded, the date and time, and the GPS position if the phone recorded one. From those, `da tag` makes:

Tag	Example	From
`maker:`	`maker:apple`	the camera's maker
`camera:`	`camera:apple-iphone-12-pro`	the camera's model
`owner:`	`owner:jane`	who owned that camera (see below)
`year:`	`year:2021`	when it was taken
`gps`	`gps`	the photo has a GPS position

1. **Make them for the whole archive:** in a terminal in the archive, `da tag` . It asks first, then works through every filed photo (photos filed before DA kept camera details are read once, roughly an hour per 50,000). To do one year or one folder: `da tag photos\organized\2021` .
2. **Say who owned each camera** (most cameras don't record it): `da tag --cameras` lists every camera found, with how many photos it took and in which years, then asks about each one with no owner yet: `Who used each camera? Type a name, or press Enter to skip. Apple iPhone 7 Plus (160 photos, 2019): jane Apple iPhone 8 (13 photos, 2019): Update the owner: tags now? [y/n] (y):` DA saves the names in the archive's da-archive.yaml for you (keeping a copy of the old file), and updates the `owner:` tags straight away. To set one without the questions: `da tag --owner "Apple iPhone 8" sam` . A camera listed with a serial number (`Nikon D90 #3012345`) is that one camera body; without one, every camera of that model.

Your own tags go on one photo, several, or every file in a folder (documents too). Name the files or folders first, then the tags:

```
da tag photos\organized\2021\2021-06 beach "big bear"
da tag photos\organized\2021\2021-06\photo_20210601_100000_ab12.heic grandma
da tag photos\organized\2021\2021-06 --remove beach
```

Tags are kept in lower case, with spaces as `-` (`big-bear`). Making the automatic tags again never changes yours.

Switch between archives

- **6 Switch archive** in the menu lists every archive on this computer. Type its number.
- Or double-click that archive's `Open DA.cmd`.
- Or in a terminal: `da archives` lists them, and `da archives --open 2` opens number 2.

How DA knows your archives: each computer keeps a list, one archive folder per line, in `%LOCALAPPDATA%\DA\archives.txt` (on computers that ran DA's older versions, `%LOCALAPPDATA%\MindIt\archives.txt`). `da setup` adds a new archive to it, and so does opening an archive (its `Open DA.cmd`, or a command run inside it), so an archive on a USB drive joins the list the first time you open it on that computer. An archive whose folder is gone (a drive that isn't plugged in) is left out of the list until it's back.

Update DA

- **DA installed with Python:** type `da update`. That's all.
- **DA.exe:** download the newest DA.exe and install it the same way as the first time (Getting started). Close any open DA windows first.

Your archives aren't touched either way. To see which version you have: `da version` (and `da update` tells you how to update the kind you have).

Less common workflows

Workflow	What it's for
iPhone Live Photos (.MOV files)	The short videos that come with iPhone photos
iPhone HEIC photos	The iPhone's photo format
Edit an archive's settings	Change its name, years or intake folders
Share an album with Google Photos	Let family see some photos online
Process a folder from somewhere else	A USB stick, an old backup, another drive
Start a new archive for another person or decade	One archive per person per decade
Use DA on another computer	Take an archive on a USB drive to another PC
Merge two copies of an archive	Two drives that each got different photos: make one whole
Back up an archive	Keep a second copy safe
Check what's already filed	A check-up of `organized`
Do the run's steps one at a time	For a careful look between each step
Location and GPS workflows	Where photos were taken; folders by place (menu **L**)
Coming soon	What DA will do next

iPhone Live Photos (.MOV files)

An iPhone *Live Photo* is two files with the same name: the photo (`IMG_1234.HEIC` or `.JPG`) and a 2-3 second video (`IMG_1234.MOV`).

- **DA keeps the pair together.** When it files the photo, the video goes with it, renamed to match: `photo_20210601_100000_ab12.heic` and `photo_20210601_100000_ab12.mov` , side by side.
- **Videos left behind by an earlier run** (filed before DA did this, or copied in after their photo was filed) can be found and put beside their photo. In a terminal, in the archive: `powershell da pair-movies # shows what it found; nothing moves da pair-movies --apply # moves them` DA matches each video with the photo that was filed from the same place under the same name. It lists what it moved, and

`da intake undo "<the file it names>"` puts them back. A video whose photo hasn't been filed yet is left alone; file the photo first.
- **Other videos** (not part of a Live Photo) stay in intake: DA doesn't file videos yet.
- **Don't want Live Photos?** On the iPhone, turn off **Live** (the round icon at the top of the camera) before taking photos.

iPhone HEIC photos

iPhones save photos as **HEIC** (`IMG_1234.HEIC`): the same quality as JPG in about half the space.

- **DA files HEIC photos like any other,** using the date the iPhone stored in them.
- **To see them on Windows,** install the free **HEIF Image Extensions** from the Microsoft Store (open the **Microsoft Store** app and search for *HEIF*).
- **To make JPG copies** (for sharing, or for a program that can't open HEIC): `powershell da heic-to-jpg photos\organized\2021\2021-06` The copies go in a new folder next to it (`2021-06-jpg`), with the same date taken; add `--out D:\to-share` to put them somewhere else. The HEIC originals stay where they are. Running it again only converts photos that are new or changed.
- **To get JPGs from the iPhone instead,** on the iPhone open **Settings > Photos** and, under *Transfer to Mac or PC*, choose **Automatic**. The iPhone then converts photos to JPG as they're copied. (**Keep Originals** copies the HEIC files.)
- **To find the HEIC files in the archive:** `da db query "SELECT path FROM files WHERE extension='.heic'"` .

Edit an archive's settings (da-archive.yaml)

Each archive's settings are in `da-archive.yaml` in its folder.

1. **Make a copy first:** right-click `da-archive.yaml` > **Copy**, then **Paste** in the same folder. If anything goes wrong, rename the copy back.
2. **Open it with Notepad** (right-click > **Open with** > **Notepad**).
3. **Change what you need.** Keep the spaces at the start of each line exactly as they are; they matter. Lines starting with `#` are notes and are ignored. ```yaml archive: name: "jane-archive-199x" # the name shown at the top of the menu years: "1990-1999" # the years this archive takes; null for any year

   intake: # the folders in intake\ and what each one holds photos: photo docs: document scans: document receipts: document # an extra folder: anything in it is filed as a document

   cameras: # who owned each camera, for owner: tags (see da tag) "Apple iPhone 12 Pro": jane ``` 4. **Save, then check:** open the archive's menu. If DA can't read the file, it says which line is wrong; fix it or put the copy back.

A new intake folder you add here is created the next time you run `da setup` on the archive (`da setup D:\jane-archive-199x`), or you can create it yourself.

Share an album with Google Photos

DA doesn't upload to Google (Google no longer lets apps like DA create shared albums), but your filed photos are easy to share by hand:

1. Go to photos.google.com and sign in.
2. Click **Upload** (or drag photos onto the page) and pick them from the archive, e.g. `D:\jane-archive-199x\photos\organized\1998\1998-07\` .
3. Click **Create** (the **+**) > **Album**, give it a name, and add the uploaded photos.
4. In the album, click **Share** and choose people, or **Create link** to send the link to anyone.

The originals stay in your archive; Google gets copies.

Process a folder from somewhere else

For photos on a USB stick, an old backup drive, or anywhere outside the archive:

```
da run E:\old-backup\photos --archive D:\jane-archive-199x
```

DA offers to **copy** the folder into the archive's `intake\` and processes the copy, so it can check every photo against what's already filed. The originals aren't touched; delete them yourself once you're happy.

Start a new archive for another person or decade

```
da setup D:\sam-archive-201x --years 2010-2019
```

Then double-click its `Open DA.cmd` . To make an archive from a folder that already has photos in it, point `da setup` at that folder: DA moves its contents into `intake\` , and lists every move in `output\runs\<date>-setup\moved.csv` (where each item was, and where it went).

Use DA on another computer

An archive is just a folder, so it can live on a USB drive and move between computers.

1. Install DA on the other computer.
2. Plug in the drive and double-click the archive's `Open DA.cmd` . It works whatever drive letter Windows gives the drive.

3. Each computer keeps its own list of archives for **6 Switch archive**. Opening an archive once with its `Open DA.cmd` adds it to that computer's list.

Merge two copies of an archive

Two copies of the same archive on two drives (say `D:\jane-archive-201x` and `E:\jane-archive-201x`) can drift apart: photos added to one, not the other. `da merge` makes one of them whole.

1. **Open the one to keep** (its `Open DA.cmd`), the one that should end up with everything.
2. **See what it would copy:** `da merge --from E:\jane-archive-201x`. DA indexes this archive, compares the other one file by file (by content, not by name), and says: `48,210 files in jane-archive-201x: 47,903 already here. Copy 307 files (2.1 GB) into jane-archive-201x: 291 to the same folder and name here 16 to intake\...\from-jane-archive-201x (their intake; da run files them)`
3. **Press Enter** (or add `--yes`) to copy them. Each file comes with its sidecar.
4. If some went to intake, choose **1 Process intake** to file them.

Good to know: - **The other archive is never changed.** DA only reads it. When you're happy with the merge, you can keep it as a backup or delete it. - **Duplicates aren't copied.** A photo this archive already has, under any name or in any folder, is skipped. - **Where copies go:** a filed photo goes to the same `YYYY\YYYY-MM\` folder and name it had there (location folders too). It goes to `intake\photos\from-<other archive>\` instead, for `da run` to file, when it's from a year outside this archive's years, when a different photo already has that name here, or when the two archives name files differently. - **Changed your mind?** DA prints an **Undo** line: `da intake undo "<the file it names>"` removes the copies again (only copies that haven't changed since). - DA checks there's room on the drive first.

Back up an archive

Everything about an archive is inside its folder: the photos, the documents, the settings, the sidecar notes and the records. To back it up, copy the **whole archive folder** to another drive (File Explorer: right-click the folder > **Copy**, open the backup drive, **Paste**).

- `_duplicates-holding\` only holds copies; you can leave it out.
- The `.da\` folder holds DA's index. It's fine to copy it, and DA can rebuild it if it's missing (**5 Refresh index**).

Check what's already filed

```
da validate photos\organized
da validate documents\organized\2010
```

This checks names, folders and sidecar notes, and finds duplicates inside that folder. It's a check only: nothing moves (it just adds any missing sidecar notes, and remakes any contact sheet whose folder has changed). The report is saved in `output\validateit\`.

To fix what it found in `organized` (wrong names, a photo in the wrong month), add `--mode repair`. On intake, or on a folder outside the archive, repair stops and tells you to use da run instead, which sets duplicates aside first.

Do the run's steps one at a time

The same steps as da run, one command each, for when you want to look closely between them:

```
da db build                                          # 1 index
da intake hash --folder phone-2024                   # 2 hash (folder inside intake)
da intake set-aside --folder phone-2024              # 3 plan only: what would be set aside
da check output\runs\<ts>-set-aside\plan.csv         #   re-check a sample of the plan
da intake set-aside --folder phone-2024 --apply      #   set them aside
da validate intake\phone-2024 --kind photo           # 4 preview
da check output\validateit\validateit-report-<ts>.json   # 5 check
da validate intake\phone-2024 --kind photo --mode repair --force   # 6 file (--force: you did 2
da verify <the repair report it printed>
da db build                                          # 7 refresh
```

Coming soon

- **Videos:** file `.mov`, `.mp4` and older videos by the date inside them, like photos.
- **Faces:** `da face-recognition` - recognise the people in photos and tag them by name.

Location and GPS workflows

Most phones record **where** each photo was taken (its GPS position); older cameras usually don't. DA reads the position when it first files a photo (or when you run da tag on photos filed earlier) and keeps it in the photo's sidecar.

Works today

- **See which photos have a position:** `da tag` (reads it from each photo once), then `da findit tags photos\organized -r gps -v`.

- **Find the ones without:** `da findit tags photos\organized -x gps -v`.
- **Put the position in file names:** use `{gps_compact}` (`N39W89`) or `{gps_lat_dir:2}{gps_lon_dir:2}` (`N39.78W89.65`) in the name pattern; see Config tokens.

Look up where photos were taken

`da location lookup` turns each photo's position into a town (or city) and state, and tags the photo with it: `place:springfield-il`, `state:il`, `country:us`.

1. Open a terminal in your archive (double-click `Open DA.cmd`).
2. Type `da location lookup` for all your filed photos, or name a folder: `da location lookup photos\organized\2019`.
3. DA says how many photos have a position and how many new places it has to look up, about one a second (1,000 new places: about 17 minutes). Press Enter to start.
4. When it's done it lists the places it found, with how many photos each.

Good to know: - Photos close together (within about 100 m) share one lookup, and every lookup is remembered in the archive, so running it again later only looks up new photos. Run it after each `da run`. - Photos filed before DA read camera details need `da tag` first; DA tells you if none have a position. - No internet? The photos it couldn't look up just wait; run `da location lookup` again later. `da location lookup --offline` uses only the places already remembered. - Photos taken at sea or in the wild may have no town; DA says how many.

Then find photos by place: - `da findit tags photos\organized -r place:springfield-il -v` (by town), `-r state:il` (by state), `-r country:fr` (by country). - Or ask the index: `da db query "SELECT path FROM files WHERE city = 'Springfield'"`.

Group a month into location folders

Instead of one big `2019-04` folder, put each outing in its own folder, named by you and by where it was: `2019\2019-04 Easter at grandma's (springfield-il)\`.

1. In the menu choose **L Location folders**. DA lists the months that have GPS photos; type a number. (Or type `da location group photos\organized\2019\2019-04`.)
2. DA looks up any places it doesn't know yet, then shows the **groups**: photos taken within 6 hours and about 2 km of the one before belong together, so a day out, or a weekend in one place, is one group. Each line shows how many photos, when, and where: `1 24 photos Sat 20 Apr 10:00 - Sat 20 Apr 16:40 springfield-il (24) 2 9 photos Sat 27 Apr 15:00 chicago-il (9)`
3. Type a **title** for each group you want a folder for, like `Easter at grandma's`. Press Enter to leave a group in the month folder. Give two groups the **same title** to put them in one folder.
4. DA shows the folder names it will make, using your folder pattern, and asks before it moves anything. Press Enter to go ahead.

5. The photos move, with their sidecars and any raw files or Live Photo videos, and each new folder gets its contact sheet. DA prints an **Undo** line: to put everything back, type `da intake undo "<the file it names>"`.

Good to know: - Photos without GPS stay in the month folder, unless you give them a place first. If the month folder ends up empty, DA removes it. - Groups too big or too small? `--hours 3` or `--km 0.5` makes smaller groups; `--hours 12 --km 10` makes bigger ones: `da location group photos\organized\2019\2019-04 --hours 3`. - `da validate` and `da run` leave photos in a location folder of their own month. - **New photos later?** `da run` files them in the plain month folder. Choose **L** on that month again: a group taken at the same time and place as a folder you already made says `fits the folder already there`, and pressing Enter adds the photos to it (type `-` to leave them). - Without the menu, give the titles in the command: `da location group photos\organized\2019\2019-04 --title "1=Easter at grandma's" --title "2=Zoo"`. A title that is an existing folder's title adds the photos to that folder.

Rename a location folder

Typed the wrong title, or thought of a better one?

```
da location rename "photos\organized\2019\2019-04 Easter (springfield-il)" "Easter at grandma's
```

DA shows the new name, `2019-04 Easter at grandma's (springfield-il)`, and asks first. The month and places in the name stay; the title changes in the folder name, in every photo's sidecar, in the index and on the contact sheet. It prints the command that renames it back.

Set a place by hand

For photos with no GPS (an old camera, a scan of a print), or when the looked-up place is wrong:

- **By name:** `da location set "photos\organized\2019\2019-04\photo_20190420_150000_ab12.jpg" --place "Springfield, IL"`. DA looks the name up, shows what it found, and asks first. A town and state, or a town and country (`"Paris, France"`), works best.
- **Same as another photo:** `--same-as <a photo from that day>` copies its place. Nothing is looked up.
- **A whole folder:** name the folder instead of a photo.
- **Take it off again:** `da location set <photos> --clear`. The next `da location lookup` looks the photo up again (if it has GPS).

A place you set is never replaced by a lookup. Photos without GPS that have a place join a group when you group the month, as if taken at that place.

Privacy: looking up places sends each spot's position (rounded to about 100 m, never the photo or its name) to OpenStreetMap's free lookup service, Nominatim. To keep positions on this computer, set `lookup: off` under

`location:` in `da-archive.yaml`:

```
location:
  lookup: off
```

Config tokens

A **token** is a word in curly brackets, like `{hash}`, that DA replaces with something about the photo. Tokens make up the patterns DA names files (and, soon, folders) with. The patterns are in the archive's da-archive.yaml:

```
modules:
  photoit:
    pattern: photo_{date_taken:%Y%m%d_%H%M%S}_{hash:4}      # photos
  scanit:
    pattern: doc_{date_taken:%y%m%d_%H%M%S}_{hash:4}        # documents
```

File-name tokens

Token	Example	What it is
`{date_taken:%Y%m%d_%H%M%S}`	`20190420_170509`	when it was taken, written with the date codes below
`{year}` `{month}` `{day}`	`2019` `04` `20`	parts of the date
`{hash}` / `{hash:4}`	`ab12cd34...` / `ab12`	the photo's fingerprint (`:4` = its first 4 characters)
`{orig_name}` / `{orig_stem}` / `{orig_ext}`	`IMG_1234.HEIC` / `IMG_1234` / `heic`	its name before DA renamed it
`{orig_folder}` / `{orig_folder:10}`	`Easter 2019` / `Easter 201`	the folder it came from (`:10` = first 10 characters)
`{tag}` / `{tags}`	`beach` / `beach_grandma`	your first tag / all your tags (da tag)
`{gps_compact}`	`N39W89`	where it was taken, whole degrees
`{gps_lat_dir:2}{gps_lon_dir:2}`	`N39.78W89.65`	the same, to 2 decimal places
`{gps_lat:4}` / `{gps_lon:4}`	`39.7817` / `-89.6501`	the position as plain numbers

Date codes for `{date_taken:...}`:

Code	Example		Code	Example
`%Y`	`2019`		`%d`	`20` (day)
`%y`	`19`		`%a` / `%A`	`Sat` / `Saturday`
`%m`	`04` (month)		`%H` `%M` `%S`	`17` `05` `09` (hour, minute, second)
`%b` / `%B`	`Apr` / `April`		`%j`	`110` (day of the year)

Examples for a photo taken on 20 April 2019 at 17:05:09, first named `Easter 2019\IMG_1234.HEIC`, tagged `beach` and `grandma`, taken at 39.78 N, 89.65 W:

Pattern	File name
`photo_{date_taken:%Y%m%d_%H%M%S}_{hash:4}` (the default)	`photo_20190420_170509_ab12.heic`
`{date_taken:%Y%m%d}_{hash:8}`	`20190420_ab12cd34.heic`
`{date_taken:%Y-%m-%d_%H%M%S}`	`2019-04-20_170509.heic`
`{date_taken:%Y-%m-%d}_{orig_stem}`	`2019-04-20_img_1234.heic`
`{orig_stem}_{date_taken:%Y%m%d}`	`img_1234_20190420.heic`
`{year}-{month}-{day}_{orig_folder}_{hash:4}`	`2019-04-20_easter_2019_ab12.heic`
`{date_taken:%Y-%m-%d (%a) %H.%M}_{hash:4}`	`2019-04-20_(sat)_17.05_ab12.heic`
`{date_taken:%Y%m%d}_{tag}_{hash:4}`	`20190420_beach_ab12.heic`
`{date_taken:%Y%m%d_%H%M%S}_{tags}`	`20190420_170509_beach_grandma.heic`
`{date_taken:%y%m%d_%H%M}_{gps_compact}`	`190420_1705_N39W89.heic`
`{date_taken:%Y-%m-%d}_{gps_lat_dir:2}{gps_lon_dir:2}_{hash:4}`	`2019-04-20_N39.78W89.65_ab12.heic`
`{date_taken:%Y%m%d}_{gps_lat:4}_{gps_lon:4}`	`20190420_39.7817_-89.6501.heic`

Rules: - Names are made lower case and spaces become `_` (compass letters like `N39W89` stay capitals). - A token with nothing to show (no tags, no position) disappears, with its extra `_`. - Keep `{hash}`, or the time to the second, in the pattern: without it two photos can end up with the same name. - **Choose before you file.** Changing the pattern later makes every photo already filed count as misnamed, and the next repair renames them all.

Folder tokens

`da run` files photos in `organized\YYYY\YYYY-MM\`. Location folders (see Group a month into location folders) are named by a folder pattern, set in `da-archive.yaml`:

```
location:
  folder: "{YYYY}-{MM} {title} ({place})"      # the default
```

Token	Example	What it is
`{YYYY}` `{MM}` `{DD}`	`2019` `04` `20`	the date of the group's first photo
`{title}`	`Easter at grandma's`	what you type for the group (kept as typed)
`{street}`	`maple-street`	the road
`{neighbourhood}`	`old-town`	
`{city}`	`springfield`	city, town or village
`{county}`	`sangamon-county`	
`{state}` / `{state:2}`	`illinois` / `il`	`:2` = the 2-letter code
`{country}` / `{country:2}`	`united-states` / `us`	
`{place}`	`springfield-il`	short for `{city}-{state:2}` (outside the US, `{city}-{country:2}`)

Folder pattern	Folder name
`{YYYY}-{MM} {title} ({place})` (the default)	`2019-04 Easter at grandma's (springfield-il)`
`{YYYY}-{MM} {title} ({street}-{city}-{state:2})`	`2019-04 Easter at grandma's (maple-street-springfield-il)`
`{YYYY}-{MM} ({city}-{state:2})`	`2019-04 (springfield-il)`
`{YYYY}-{MM}-{DD} {title}`	`2019-04-20 Easter at grandma's`
`{YYYY}-{MM} {title} ({city}, {state})`	`2019-04 Easter at grandma's (springfield, illinois)`
`{YYYY}-{MM} {title} ({city}-{country:2})`	`2019-07 Summer trip (paris-fr)`
a group in more than one place	`2019-04 Spring trip (springfield-il, chicago-il)`
a group with no position	`2019-04 Easter at grandma's`

Place tokens are lower case with `-` for spaces; empty ones are dropped with their separators, and empty brackets disappear. A location folder always sits in its year folder (`organized\2019\2019-04 Easter at grandma's (...)`), and its name starts with the month, so DA knows the photos are in the right place.

You can also make one by hand. Rename a month folder (or make a new folder next to it and move some of that month's photos in, with their `.meta.json` files), keeping the month at the start: `2019-04` -> `2019-04 Easter at grandma's` . `da validate` and `da run` leave photos there. A photo from another month in that folder is still moved back to its own month folder. New photos from `da run` go into the plain month folder.

If something goes wrong

What you see	What to do
'da' is not recognized	Close the terminal and open a **new** one. If that doesn't help, install DA again.
Windows protected your PC	Click **More info**, then **Run anyway**. DA isn't signed by a company, so Windows asks once.
Not inside an archive	Open the archive with its `Open DA.cmd`, or type `da` and pick one, or add `--archive D:\jane-archive-199x` to the command.
DA.exe is in use while updating	Close every DA window, then install again.
A run seems stuck	It's probably working through a big folder. Big runs take hours; the step number shows where it is. To stop, press **Ctrl+C**. Photos already filed stay filed, and running it again carries on with the rest.
A photo was filed with the wrong date	The camera's clock was wrong. Rename the file in `organized` with the right date at the start, and move it to the right month folder.
Nothing to file	Everything in that folder is either filed or waiting for you. See **3 What needs review**.
Files are in `_duplicates-holding` that you want back	**4 Undo last set-aside**, or `da intake undo "<run folder>\set-aside-moved.csv"`.
You want to know what DA did, and when	Every command is written to the archive's log, `logs\da.jsonl`: when it started, how it ended, what it moved, with the undo file for each move. Errors are there with the full details, which helps when you ask for help.

Reference

How DA works out a date

In this order:

1. A date **and time** in the file name (`2024_07_04_10_15_00.jpg` , `20240704_101500.jpg`).
2. The date stored **inside** the file:
    - photos: the camera's "date taken", including iPhone HEIC photos. A camera whose clock was never set (1 January 1970) doesn't count.
    - PDFs: when the PDF was made (scanned, printed or saved).
    - Word, Excel, PowerPoint: when the document was created.
3. A date **without a time** in the name (`2010-04-27` , or US-style `04-27-2010`). If the date inside the file is from that same day, its time is used too.
4. A **month and year** in the name (`Resume April-2010`): the date inside the file, if it's from that month. Otherwise only the month is known, which isn't enough, and the file stays for review.
5. A **year only** (`Beach 1998.jpg`): not enough; it stays for review.

When the name and the date inside disagree, the name wins, and the file is listed for review.

Photos are named `photo_YYYYMMDD_HHMMSS_xxxx.jpg` and documents `doc_YYMMDD_HHMMSS_xxxx.pdf` ; the last four characters come from the file's fingerprint, so two photos taken in the same second still get different names.

Photo or document?

PDFs and Word, Excel and PowerPoint files are always documents. A picture (`.jpg .png .tif .bmp`) could be a photo or a scan, so the intake folder decides: in `docs\` or `scans\` it's a document; anywhere else it's a photo. To decide for a whole run: `da run intake\old-scans --kind document` .

Which archive a command works on

1. The one you name: `--archive D:\jane-archive-199x` (before or after the command).
2. Otherwise, the archive the folder you name is in.
3. Otherwise, the archive you're in: the terminal's current folder, or one above it.

Commands

Command	What it does
`da`	The menu
`da run [folder]`	File what's in intake (or one folder of it)
`da status`	This archive at a glance
`da report [--open] [--refresh]`	A dated one-page PDF about the archive, in its `output` folder
`da setup <folder> [--years 1990-1999]`	Create an archive, or turn a folder into one
`da archives [--open N]`	The archives on this computer
`da guide [--pdf]`	Open this guide (or its PDF)
`da db build` / `da db stats` / `da db query "<SQL>"`	The index: refresh, counts by year, search
`da intake status` / `da intake undo <moved.csv>`	Intake progress; put duplicates back
`da validate <folder>`	Check a folder (nothing moves)
`da contact-sheets <folder> [--force]`	Make the contact sheets for every folder of photos in it
`da tag [files or folders] [tags] [--remove tag]`	Automatic tags, or add or remove your own
`da tag --cameras`	The cameras found; asks who used each and saves it
`da tag --owner "<camera>" <name>`	Save who used a camera
`da findit tags <folder> -r <tag>`	Find files by tag
`da location lookup [folder] [--offline]`	Look up the town and state of GPS photos; adds place tags
`da split [--to <archive>] [--years 2000-2004]`	Move files from other years to the archive for them

Command	What it does
`da merge --from <other archive>`	Copy in what another copy of the archive has and this one doesn't
`da location group <month folder>`	Group a month's GPS photos into location folders (menu **L**)
`da location rename <folder> "<title>"`	Give a location folder a new title
`da location set <photos> --place "<town, state>"`	Set a place by hand (`--same-as <photo>` , `--clear`)
`da pair-movies [--apply]`	Put Live Photo videos left in intake beside their filed photo
`da heic-to-jpg <folder> [--out <folder>]`	JPG copies of HEIC photos
`da date list` / `da date set <photos> --date 1998-07` / `da date shift <photos> --by +1h`	Fix dates (menu **D**)
`da version`	Which version of DA this is
`da update`	Update DA (installed with Python: does it; DA.exe: says how)
`<DA.exe> install`	Install or update DA on this computer